

Interoperability Framework

The digital home will consist of a network of CE, mobile, and PC devices that cooperate transparently, delivering simple, seamless interoperability that enhances and enriches user experiences. This network is the communications and control backbone for the digital home and is based on IP networking and UPnP technology. Interoperability is accomplished between devices when they are capable of transparently collaborating on a particular service that they provide to the user. Typically, this includes the capability for these devices to communicate with each other and exchange meaningful information. The digital home building blocks needed to facilitate this interoperability are described below.

- m Transparent connectivity between devices inside the digital home: Includes networking compatibility at the link layer (layer 2) for devices directly connected to each other. When devices of different layer two technologies need to communicate, appropriate layer 2 bridging and layer 3 routing must exist between these devices. The overall goal is to enable end-to-end connectivity between all devices exchanging information over the home network.
- m Unified framework for device discovery, configuration and control: The ability for any device on the digital home network to discover the presence of other devices and services on the network, and identify their functionality and associated capabilities. It also includes the ability to configure these devices and services, and control their operation with appropriate ease-of-use.
- m Interoperable media formats and streaming protocols: Once devices can communicate with each other, they need to agree on a common streaming protocol in order to establish media streaming sessions. These devices also need to agree on the media formats that they support to ensure that the media can be shared and consumed.
- m Interoperable media management and control framework: An interoperable media management framework across all devices in the digital home enables the proper exchange of media information and control between devices provided by different vendors. It must include the ability to organise, browse, search, and select media items to be processed, in addition to the ability to control the operation of media streaming sessions.